

Constraining Planet Occurrence in the Milky Way: The Missing Role of Dynamics

Stars and their planetary systems migrate many kiloparsecs over their lifetimes; we can directly distinguish between stars that have radially migrated from those that have not through observable kinematic signatures (i.e. eccentricity and orbital size). Our Sun has migrated several kiloparsecs from its birth radius; therefore, it is critical to analyze radial migration histories when it comes to searching for Sun-like stars for habitability missions. Previous studies have shown that host star metallicity exhibits weak correlations with planet occurrence. This points to a degeneracy: stars with similar metallicities can have very different radial migration histories. However, to break this degeneracy, we need a framework that connects the radial migration histories of stars with observed planet occurrence rates. We propose to (1) build cutting-edge tracer particle simulations to identify kinematic signatures of host stars that have experienced strong dynamical encounters with spiral arms and (2) reveal how these perturbations correlate with planet outcomes. PLATO will deliver high-resolution observational data, while existing computing resources will enable the build of precision simulations. Comparing these populations within fixed metallicity ranges will help break the degeneracy between metallicity and radial migration history. Understanding whether there are dynamical differences between stars in the Milky Way would redefine how we inform target selection for next-generation habitability missions. The results of this work are critical to PLATO's exoplanets program and broader efforts to understand the correlation between planets and their host stars.

- **Scientific Justification**

Radial Migration of Host Stars is the Missing Key to Planet Habitability

Understanding the radial migration history of a host star is critical and currently a missing parameter in determining planet occurrence. Stars and their planetary systems migrate several kiloparsecs over their lifetimes (Roškar+2012); the Sun, a star we know to host life, has moved significantly from its birth radius (Wielen+1996). We want to understand whether radial migration of stars relates to its ability to host planets. We can directly distinguish between stars that have radially migrated from those that have not through observable kinematic signatures (i.e. eccentricity and orbital size). Previous studies have shown only weak correlations between host star metallicity and planet occurrence; this reflects a fundamental degeneracy: stars with similar metallicities can have very different radial migration histories (Sayeed+2025).

However, to break this degeneracy, we need a framework that connects the radial migration histories of stars with observed planet occurrence rates. This is not only timely but highly feasible: PLATO will deliver high-resolution observational data, while existing computing resources will enable the build of precision simulations.

We propose to (1) build cutting-edge tracer particle simulations to identify kinematic signatures of host stars that have experienced strong dynamical encounters with spiral arms and (2) reveal how these perturbations correlate with planet outcomes. From our simulation outputs, it will be possible to identify kinematic signatures of stars that have experienced dynamical perturbations. PLATO will allow us to observationally follow-up and identify whether a star being perturbed or not affects the number of planets it hosts. Comparing these populations within fixed metallicity ranges will help break the degeneracy between metallicity and radial migration history. The results of this work will directly support ongoing PLATO's exoplanet program and contribute to broader efforts to understand the correlation between planets and their host stars. Understanding whether there are dynamical differences between stars in the Milky Way would redefine how we inform target selection for next-generation habitability missions.

We Lack a Framework Connecting Radial Migration of Host Stars to Planet Occurrence

We can directly distinguish between stars (Smock+in prep.) that have radially migrated from those that have not through observable kinematic signatures (i.e. eccentricity and orbital size). Previous studies (Sayeed+2025) have shown that host star metallicity exhibits only weak correlations with planet occurrence, yet stars with similar metallicities can have vastly different radial migration histories. This points to a fundamental degeneracy: chemistry alone is insufficient to predict planet outcomes. However, to break this degeneracy, we need a framework that connects the radial migration histories of stars with observed planet occurrence rates. Our Sun, a star we know to host life, has migrated several kiloparsecs from its birth radius (Wielen+1996); therefore, it is

critical to analyze radial migration histories when it comes to searching for Sun-like stars for habitability missions. Understanding whether dynamical history of host stars drives variation in planet occurrence would fundamentally redefine how we inform target selection for exoplanet surveys.

It is well established that stars in the Milky Way’s thin and thick disks exhibit distinct kinematic signatures, suggesting differences in their dynamical origins (Sagear+2025). We propose to (1) build cutting-edge tracer particle simulations to identify kinematic signatures of host stars that have experienced strong dynamical encounters with spiral arms and (2) reveal how these perturbations correlate with planet outcomes. It is also well established in the literature that assuming radial migration in tracer particle simulations is an accurate model of stellar dynamics in the Milky Way. However, no study has taken this a step further by combining this approach with observational data from exoplanet surveys such as PLATO to test whether host star dynamics are linked to planet occurrence.

This Proposal: Leveraging ESA’s PLATO Mission to Link Radial Migration Histories with Planet Occurrence

We aim to combine tracer particle simulations with observations from ESA’s PLATO mission to determine whether radial migration of host stars influences planet occurrence in the Milky Way. It is not simply the detection of exoplanets that we require, but a large, homogeneous sample of host stars with both precise stellar properties and well-characterized planetary systems, capabilities uniquely enabled by ESA. Only ESA’s PLATO can provide the dataset necessary to connect the radial migration histories of host stars with planet occurrence.

We propose to (1) build cutting-edge tracer particle simulations to identify kinematic signatures of host stars that have experienced strong dynamical encounters with spiral arms and (2) determine how these perturbations correlate with planet occurrence (*see* Fig. 1). The resulting simulations alongside HDBSCAN (McInnes+2017) will allow us to classify stars as (i) perturbed or (ii) unperturbed. We will then use the Gaia IDs of these selected stars to cross-match with PLATO observations and determine whether stars in each dynamical group exhibit differences in planet occurrence rates (Smock+in prep.).

We will build a high-resolution tracer particle simulation to capture the radial migration histories of stars and deliver a framework that connects these simulations with PLATO data to link stellar dynamics to planet occurrence. With ESA, we can create a never-before-developed framework relating the dynamics of stars to exoplanets and transform how we identify habitable worlds.

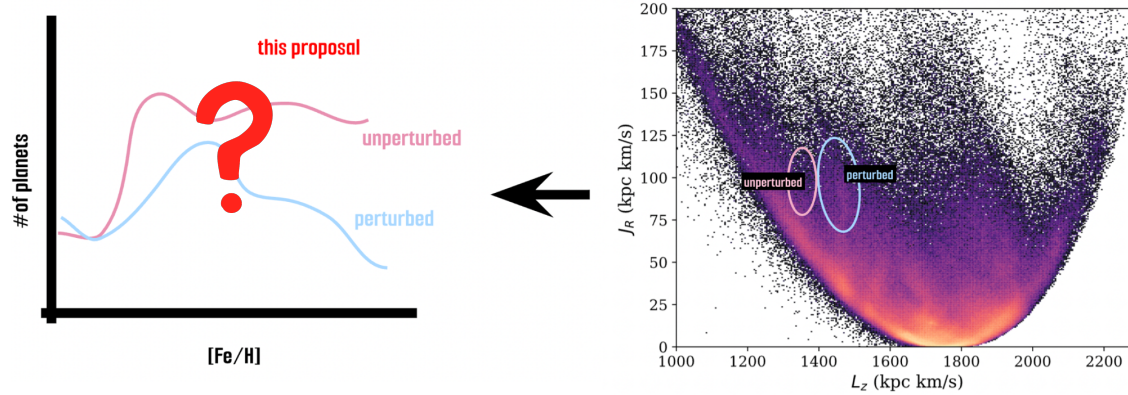


Fig 1. Radial migration of stars is the missing key to understanding planet occurrence in the Milky Way. The *right* panel shows the distribution of $\sim 7 \times 10^5$ stars within 200 pc of the Sun in action space (J_R , L_z), combining 6D kinematics from Gaia DR3, GALAH DR3, and APOGEE DR16 (Rampalli+2023). The diagonal over-densities (“wrinkles”) in this space trace dynamical perturbations, particularly at high J_R (Smock et al., in prep.); the blue oval marks perturbed stars and the pink oval marks unperturbed stars. The *left* panel illustrates the core problem: at fixed metallicity ($[Fe/H]$), we aim to determine whether stellar dynamics alone (perturbed vs. unperturbed) drives differences in planet occurrence. Stars and their planetary systems migrate many kiloparsecs over their lifetimes, carrying with them the chemical fingerprints of their birthplaces; however, metallicity alone is insufficient to explain planet occurrence, and we must account for the effects of radial migration. This program will (1) build tracer particle simulations to make predictions about the radial migration histories of stars and (2) investigate how these histories correlate with planet outcomes. Understanding the impact of radial migration of stars and their planet systems is the missing key to finding exoplanets in our search for habitability.

Radial Migration of Host Stars Is Critical for Target Selection in Future Exoplanet Missions

Understanding whether radial migration of host stars influences planet occurrence is essential for advancing exoplanet science in a Galactic context. By building tracer particle simulations to analyze dynamical perturbations and then using PLATO to connect stellar dynamics to planet occurrence, we will establish a new framework that transforms how we search for habitable worlds.

ESA’s PLATO mission is launching soon and will provide an unprecedented, homogeneous dataset of exoplanets and their host star properties, making it possible for the first time to test whether radial migration histories of host stars influence planet occurrence. This program directly aligns with ESA’s scientific goal of “understanding the origin of planetary systems and the potential for life beyond Earth” by connecting host star dynamics to planet occurrence using PLATO observations. By combining PLATO data alongside tracer particle simulations into a unified, publicly available framework, this work increases the legacy value of ESA and advances how we identify stars in the search for habitable worlds, guided by the principle: “know thy star, know thy planet.”

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